

AMENDMENT NO. 7 JANUARY 2026

TO

IS 9523 : 2000 DUCTILE IRON FITTINGS FOR PRESSURE PIPES FOR WATER, GAS AND SEWERAGE — SPECIFICATION

(First Revision)

(Page 4, clause 11.2) — Insert the following at the end:

‘11.3 Fitting of DN 40, DN 50, DN 60 and DN 65 or combination of these diameter with other diameters, may be manufactured for fixing of valves and appurtenances, provided they follow the requirements of this standard. The basic dimension of such fitting are given in Table 21 and Table 29 which are modified accordingly.’

(Page 29, Annex D) — Insert the following new Annex E after Annex D:

ANNEX E

Clause 11.1

‘Dimensions of standard flange drilling for dimensions DN 40, DN 50, DN 60 and DN 65

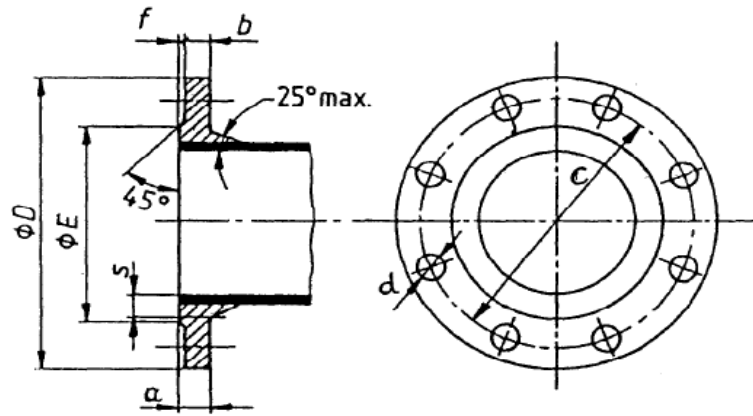


Table 35 Dimensions of Standard Flange Drilling for Dimensions DN 40, DN 50, DN 60 and DN 65

Dimensions of Standard Flange Drilling for Flange Fittings- PN 10/PN 16										
Sl No.	Nominal Diameter	Dimensions						Holes		Bolt Size Metric
		D	E	C	B	f	a	No.	Dia (d)	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
(i)	40	150	84	110	16	3	19	4	19	M 16
(ii)	50	165	99	125	16	3	19	4	19	M 16
(iii)	65	185	118	145	16	3	19	4	19	M 16

Dimensions of Standard Flange Drilling for Flange Fittings -PN 25/PN 40										
Sl No.	Nominal Diameter	Dimensions						Holes		Bolt Size Metric
		D	E	C	B	f	a	No.	Dia (d)	
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
(i)	40	150	84	110	16	3	19	4	19	M 16
(ii)	50	165	99	125	16	3	19	4	19	M 16
(iii)	65	185	118	145	16	3	19	8	19	M 16

Amendment No. 7 to 9523 : 2000

[Page 9, Table 12, col (2), title] — Substitute the existing heading for col (2) as under:

‘Allowable difference between minor axis and DE (minimum) in mm’.

(Page 9, clause 12.3.1) — Substitute the following for the existing:

Fittings shall be as far as possible circular internally and externally. The tolerance for out of roundness of spigot ends in the jointing zone for push-on-joint as given in Table 12 and for mechanical joint as given in Table 13.

(Page 9, Table 13) — Substitute the existing referred clauses as under:

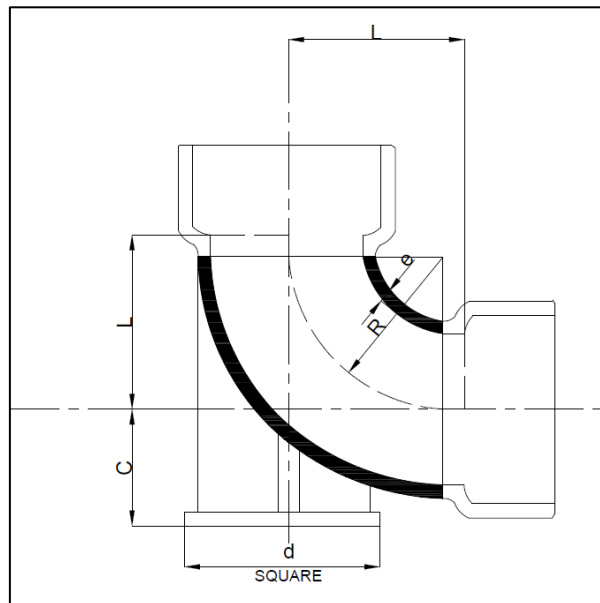
(Clause 12.3.1) for (Clauses 12.3.1 and 12.4.3)

(Page 9, clause 12.4) — Substitute the following for the existing:

The tolerance on the wall thickness of fittings shall be as follows:

<i>Type of Castings</i>	<i>Wall Thickness mm</i>	<i>Tolerance mm</i>
Fittings	7 and Greater than 7	– (2.3 + 0.001 DN)

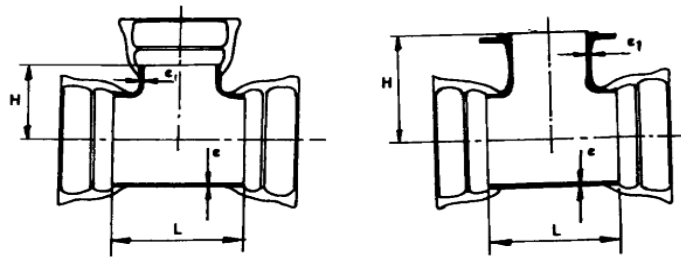
(Page 13, Table 19) — Substitute the following revised sketch for existing sketch in the table:



(Page 14, Table 21) — Substitute the following for the existing:

Table 21 All Socket, Flanged On Double Socket Tees

(Clauses 11.2 and 16)



NOTES

1 The figure is schematic.

2 No value is specified for radius R but internal and external radius are equal.

Sl No.	Nominal Dia (DN)		e	e ₁	L		H	
	Body	Branch			Branch Flange on Double Socket	All Socket	Flange on Double Socket	All Socket
(1)	mm (2)	mm (3)	mm (4)	mm (5)	mm (6)	mm (7)	mm (8)	mm (9)
(i)	40	40	7	7	120	—	130	—
(ii)	50	50	7	7	130	—	140	—
(iii)	60	40	7	7	145	—	150	—
(iv)	60	60	7	7	145	—	150	—
(v)	65	40	7	7	150	—	150	—
(vi)	65	65	7	7	150	—	150	—
(vii)	80	50	7	7	170	—	165	—
(viii)	80	60	7	7	170	—	165	—
(ix)	80	80	7	7	170	170	165	85
(x)	100	50	7.2	7	170	—	175	—
(xi)	100	80	7.2	7	170	170	175	95
(xii)	100	100	7.2	7.2	190	190	180	95
(xiii)	125	50	7.5	7	170	—	190	—
(xiv)	125	60	7.5	7	170	—	190	—
(xv)	125	80	7.5	7	170	170	190	105
(xvi)	125	100	7.5	7.2	195	195	195	110
(xvii)	125	125	7.5	7.5	225	225	200	110
(xviii)	150	50	7.8	7	170	—	205	—
(xix)	150	60	7.8	7	170	—	205	—
(xx)	150	80	7.8	7	170	170	205	120

Amendment No. 7 to 9523 : 2000

Table 21 (Continued)

Sl No.	Nominal Dia (DN)		e	e_1	L		H	
	Body	Branch			Branch Flange on Double Socket	All Socket	Flange on Double Socket	All Socket
(1)	mm (2)	mm (3)	mm (4)	mm (5)	mm (6)	mm (7)	mm (8)	mm (9)
(xxi)	150	100	7.8	7.2	195	195	210	120
(xxii)	150	150	7.8	7.8	255	255	220	125
(xxiii)	200	50	8.4	7	175	—	235	—
(xxiv)	200	60	8.4	7	175	—	235	—
(xxv)	200	80	8.4	7	175	175	235	145
(xxvi)	200	100	8.4	7.2	200	200	240	145
(xxvii)	200	150	8.4	7.8	255	255	250	150
(xxviii)	200	200	8.4	8.4	315	315	260	155
(xxix)	250	50	9	7	180	—	265	—
(xxx)	250	80	9	7	180	180	265	170
(xxxi)	250	100	9	7.2	200	200	270	170
(xxxii)	250	150	9	7.8	260	260	280	175
(xxxiii)	250	200	9	8.4	315	315	290	180
(xxxiv)	250	250	9	9	375	375	300	190
(xxxv)	300	50	9.6	7	180	—	295	—
(xxxvi)	300	100	9.6	7.2	205	205	300	195
(xxxvii)	300	200	9.6	8.4	320	320	320	205
(xxxviii)	300	300	9.6	9.6	435	435	340	220
(xxxix)	350	60	10.2	7	205	—	330	—
(xl)	350	100	10.2	7.2	205	—	330	—
(xli)	350	200	10.2	8.4	325	—	350	—
(xlii)	350	350	10.2	10.2	495	—	380	—
(xliii)	400	80	10.8	7.0	185	—	355	—
(xliv)	400	100	10.8	7.2	210	—	360	—
(xlv)	400	150	10.8	7.8	270	—	370	—
(xlvi)	400	200	10.8	8.4	325	—	380	—
(xlvii)	400	300	10.8	9.6	440	—	400	—
(xlviii)	400	400	10.8	10.8	560	—	420	—

Table 21 (Concluded)

Sl No.	Nominal Dia (DN)		<i>e</i>	<i>e₁</i>	<i>L</i>		<i>H</i>	
	Body	Branch			Branch Flange on double socket	All Socket	Flange on double socket	All Socket
(1)	mm (2)	mm (3)	mm (4)	mm (5)	mm (6)	mm (7)	mm (8)	mm (9)
(xlix)	450	100	11.4	7.2	215	—	390	—
(l)	450	250	11.4	9.0	390	—	420	—
(li)	450	450	11.4	11.4	620	—	460	—
(lii)	500	100	12.0	7.2	215	—	420	—
(liii)	500	200	12.0	8.4	330	—	440	—
(liv)	500	400	12.0	10.8	565	—	480	—
(lv)	500	500	12.0	12.0	680	—	500	—
(lvi)	600	200	13.2	8.4	340	—	500	—
(lvii)	600	400	13.2	10.8	570	—	540	—
(lviii)	600	600	13.2	13.2	800	—	580	—
(lix)	700	200	14.4	8.4	345	—	525	—
(lx)	700	400	14.4	10.8	575	—	555	—
(lxi)	700	700	14.4	14.4	925	—	600	—
(lxii)	750	400	15.0	10.8	575	—	585	—
(lxiii)	750	750	15.0	15.0	985	—	635	—
(lxiv)	800	200	15.6	8.4	350	—	585	—
(lxv)	800	400	15.6	10.8	580	—	615	—
(lxvi)	800	600	15.6	13.2	1 045	—	645	—
(lxvii)	800	800	15.6	15.6	1 045	—	675	—
(lxviii)	900	200	16.8	8.4	355	—	645	—
(lxix)	900	400	16.8	10.8	590	—	675	—
(lxx)	900	600	16.8	13.2	1 170	—	705	—
(lxxi)	900	900	16.8	16.8	1 170	—	750	—
(lxxii)	1 000	200	18.0	8.4	360	—	705	—
(lxxiii)	1 000	400	18.0	10.8	595	—	735	—
(lxxiv)	1 000	600	18.0	13.2	1 290	—	765	—
(lxxv)	1 000	1 000	18.0	18.0	1 290	—	825	—
(lxxvi)	1 100	400	19.2	10.8	600	—	795	—
(lxxvii)	1 100	600	19.2	13.2	830	—	825	—
(lxxviii)	1 200	600	20.4	13.2	840	—	885	—
(lxxix)	1 200	800	20.4	15.6	1 070	—	915	—
(lxxx)	1 200	1 000	20.4	18.0	1 300	—	945	—
(lxxxii)	1 400	600	22.8	13.2	1 030	—	980	—
(lxxxiii)	1 400	800	22.8	15.6	1 260	—	1 010	—
(lxxxiv)	1 400	1 000	22.8	18.0	1 495	—	1 040	—
(lxxxv)	1 600	600	25.2	13.2	1 040	—	1 090	—
(lxxxvi)	1 600	800	25.2	15.6	1 275	—	1 120	—
(lxxxvii)	1 600	1 000	25.2	18.0	1 505	—	1 150	—
(lxxxviii)	1 600	1 200	25.2	20.4	1 740	—	1 180	—

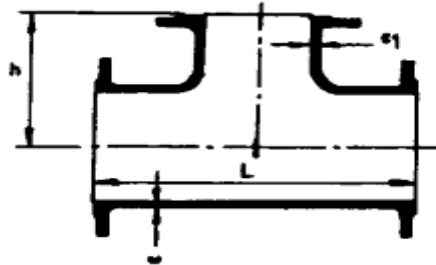
Amendment No. 7 to 9523 : 2000

(Page 21, Table 29) — Substitute the following for the existing:

Table 29

ALL FLANGED TEES

(Clauses 11.2 and 16)



SI No.	Nominal Dia (DN)		e	e_1	L	H
	Body	Branch				
	mm	mm	mm	mm	mm	mm
(1)	(2)	(3)	(4)	(5)	(6)	(7)
(i)	40	40	7.0	7.0	280	140
(ii)	50	50	7.0	7.0	300	150
(iii)	60	40	7.0	7.0	300	130
(iv)	60	60	7.0	7.0	320	160
(v)	65	65	7.0	7.0	330	165
(vi)	80	40	7.0	7.0	330	165
(vii)	80	60	7.0	7.0	330	165
(viii)	80	80	7.0	7.0	330	165
(ix)	100	40	7.2	7.0	360	175
(x)	100	60	7.2	7.0	360	175
(xi)	100	80	7.2	7.0	360	175
(xii)	100	100	7.2	7.2	360	180
(xiii)	125	40	7.5	7.0	400	190
(xiv)	125	60	7.5	7.0	400	190
(xv)	125	80	7.5	7.0	400	190
(xvi)	125	100	7.5	7.2	400	195
(xvii)	125	125	7.5	7.5	400	200
(xviii)	150	40	7.8	7.0	440	205
(xix)	150	60	7.8	7.0	440	205
(xx)	150	80	7.8	7.0	440	205
(xxi)	150	100	7.8	7.2	440	210
(xxii)	150	125	7.8	7.5	440	215
(xxiii)	150	150	7.8	7.8	440	220
(xxiv)	200	40	8.4	7.0	520	235
(xxv)	200	60	8.4	7.0	520	235
(xxvi)	200	80	8.4	7.0	520	235
(xxvii)	200	100	8.4	7.2	520	240
(xxviii)	200	150	8.4	7.8	520	250
(xxix)	200	200	8.4	8.4	520	260
(xxx)	250	60	9.0	7.0	700	275

Amendment No. 7 to 9523 : 2000

SI No.	Nominal Dia (DN)		e	e_1	L	H
	Bodv	Branch				
	mm	mm	mm	mm	mm	mm
(1)	(2)	(3)	(4)	(5)	(6)	(7)
(xxxi)	250	100	9.0	7.2	700	275
(xxxii)	250	200	9.0	8.4	700	325
(xxxiii)	250	250	9.0	9.0	700	350
(xxxiv)	300	60	9.6	7.0	800	300
(xxxv)	300	100	9.6	7.2	800	300
(xxxvi)	300	200	9.6	8.4	800	350
(xxxvii)	300	300	9.6	9.6	800	400
(xxxviii)	350	60	10.2	7.0	850	325
(xxxix)	350	100	10.2	7.2	850	325
(xl)	350	200	10.2	8.4	850	325
(xli)	350	350	10.2	10.2	850	425
(xlii)	400	100	10.8	7.2	900	350
(xliii)	400	200	10.8	8.4	900	350
(xliv)	400	400	10.8	10.8	900	450
(xlv)	450	100	11.4	7.2	950	375
(xlvi)	450	200	11.4	8.4	950	375
(xlvii)	450	450	11.4	11.4	950	475
(xlviii)	500	100	12.0	7.2	1 000	400
(xlix)	500	200	12.0	8.4	1 000	400
(l)	500	400	12.0	10.8	1 000	500
(li)	500	500	12.0	12.0	1 000	500
(lii)	600	200	13.2	8.4	1 100	450
(liii)	600	400	13.2	10.8	1 100	550
(liv)	600	600	13.2	13.2	1 100	550
(lv)	700	200	14.4	8.4	650	525
(lvi)	700	400	14.4	10.4	870	555
(lvii)	700	700	14.4	14.4	1200	600
(lviii)	750	200	15	8.4	670	555
(lix)	750	400	15	10.8	890	585
(lx)	750	750	15	15.0	1 275	640
(lxi)	800	200	15.6	8.4	690	585
(lxii)	800	400	15.6	10.8	910	615
(lxiii)	800	600	15.6	13.2	1 350	645
(lxiv)	800	800	15.6	15.6	1 350	675
(lxv)	900	200	16.8	8.4	730	645
(lxvi)	900	400	16.8	10.8	950	675
(lxvii)	900	600	16.8	13.2	1 500	705
(lxviii)	900	900	16.8	16.8	1 500	750
(lxix)	1 000	200	18.0	8.4	770	705
(lxx)	1 000	400	18.0	10.8	990	735
(lxxi)	1 000	600	18.0	13.2	1 650	765
(lxxii)	1 000	1 000	18.0	18.0	1 650	825
(lxxiii)	1 100	400	19.2	10.8	980	795
(lxxiv)	1 100	600	19.2	13.2	1 210	825
(lxxv)	1 200	600	20.4	13.2	1 240	885
(lxxvi)	1 200	800	20.4	15.6	1 470	915

Amendment No. 7 to 9523 : 2000

SI No.	Nominal Dia (DN)		e	e_1	L	H
	Body	Branch				
	mm	mm	mm	mm	mm	mm
(1)	(2)	(3)	(4)	(5)	(6)	(7)
(lxxvii)	1 200	1 000	20.4	18.0	1 700	945
(lxxviii)	1 400	600	22.8	13.2	1 550	980
(lxxix)	1 400	800	22.8	15.6	1 760	1 010
(lxxx)	1 400	1 000	22.8	18.0	2 015	1 040
(lxxxii)	1 600	600	25.2	13.2	1 600	1 090
(lxxxiii)	1 600	800	25.2	15.6	1 835	1 120
(lxxxiv)	1 600	1 000	25.2	18.0	2 065	1 150
(lxxxv)	1 600	1 200	25.2	20.4	2 300	1 180

NOTE

1 The figure is schematic.

2 No value is specified for radius R but internal and external radius are equal.

3 PN10 and PN16 are the preferred flange. PN 25 and PN 40 flanges are supplied when specified.

(Page 23, Table 30) — Substitute the following for the existing:

SI No.	Nominal dia. (DN)		e_1	e_2	L
	Larger End (Body)	Smaller End (Branch)			
(1)	(2)	(3)	(4)	(5)	(6)
(i)	50	40	7.0	7.0	150
(ii)	60	50	7.0	7.0	160
(iii)	65	50	7.0	7.0	200
(iv)	80	60	7.0	7.0	200
(v)	80	65	7.0	7.0	200
(vi)	100	80	7.2	7.0	200
(vii)	125	100	7.5	7.2	200
(viii)	150	125	7.8	7.5	200
(ix)	200	150	8.4	7.8	300
(x)	250	200	9.0	8.4	300
(xi)	300	250	9.6	9.0	300
(xii)	350	300	10.2	9.6	300
(xiii)	400	350	10.8	10.2	300
(xiv)	450	400	11.4	10.8	300
(xv)	500	400	12.0	10.8	600
(xvi)	600	500	13.2	12.0	600
(xvii)	700	600	14.4	13.2	600
(xviii)	750	600	15.0	13.2	600
(xix)	800	700	15.6	14.4	600
(xx)	900	800	16.8	15.6	600

Amendment No. 7 to 9523 : 2000

Sl No.	Nominal dia. (DN)		e_1	e_2	L
	Larger End (Body)	Smaller End (Branch)			
(1)	(2)	(3)	(4)	(5)	(6)
(xx)	1 000	900	18.0	16.8	600
(xxi)	1 100	1 000	19.2	18.0	600
(xxii)	1 200	1 000	20.4	18.0	790
(xxiii)	1 400	1 200	22.8	20.4	850
(xxiv)	1 600	1 400	25.2	22.8	910

[Page 26, Annex B, Clause B-1.1,(i)] — Substitute the following for the existing:

Ordinary Portland Cement as per IS 269 or Portland slag cement as per IS 455 to be used.